Introduction to Pandas

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`Pandas` is the central piece of the `Python` ecosystem for data science. This chapter presents the first data exploration that can be performed with `Pandas` in order to explore the structure of a dataset.

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Skills to be acquired by the end of this chapter

- Import a dataset as a `Pandas` dataframe and explore its structure;
- Perform manipulations on columns and rows;
- Construct aggregate statistics and chain operations;

- Use `Pandas` graphical methods to quickly represent data distribution.

1 Introduction

The `Pandas` package has been the central piece of the data science ecosystem for about a decade. The *DataFrame*, a central object in languages like `R` or `Stata`, had long been absent in the `Python` ecosystem. Yet, thanks to `Numpy`, all the basic components were present but needed to be reconfigured to meet the needs of data scientists.

Wes McKinney, when he built `Pandas` to provide a dataframe leveraging the numerical computation library `Numpy` in the background, enabled a significant leap forward for `Python` in data analysis, explaining its popularity in the data science ecosystem. `Pandas` is not without limitations[1], which we will have the opportunity to discuss, but the vast array of analysis methods it offers greatly simplifies data analysis work. For more information on this package, the reference book by W. McKinney [1] presents many of the package's features.

In this chapter, we will focus on the most relevant elements in the context of an introduction to data science, leaving interested users to deepen their knowledge with the abundant resources available on the subject.

As datasets generally gain value by associating multiple sources, for example, to relate a record to contextual data or to link two client databases to obtain meaningful data, the next chapter will present how to merge different datasets with `Pandas`. By the end of the next chapter, thanks to data merging, we will have a detailed database on the carbon footprints of the French[2].

1.1 Data used in this chapter

In this `Pandas` tutorial, we will use:

- Greenhouse gas emissions estimated at the municipal level by ADEME. The dataset is available on data.gouv and can be queried directly in `Python` with this URL.

The next chapter will allow us to apply the elements presented in this chapter with the above data combined with contextual data at the municipal level.

1.2 Environment

We will follow the usual conventions in importing packages:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

[1] The equivalent ecosystem in `R`, the `tidyverse`, developed by *Posit*, is of more recent design than `Pandas`. Its philosophy could thus draw inspiration from `Pandas` while addressing some limitations of the `Pandas` syntax. Since both syntaxes are an implementation in `Python` or `R` of the `SQL` philosophy, it is natural that they resemble each other and that it is pertinent for data scientists to know both languages.

[2] Actually, it is not the carbon footprint but the **national inventory** since the database corresponds to a production view, not consumption. Emissions made in one municipality to satisfy the consumption of another will be attributed to the former where the carbon footprint concept would attribute it to the latter. Moreover, the emissions presented here do not include those produced by goods made abroad. This exercise is not about constructing a reliable statistic but rather understanding the logic of data merging to construct descriptive statistics.

To obtain reproducible results, you can set the seed of the pseudo-random number generator.

```
import numpy as np
np.random.seed(123)
```

Throughout this demonstration of the main `Pandas` functionalities, and in the next chapter, I recommend regularly referring to the following resources:

- The official `Pandas` documentation, especially the language comparison page, which is very useful;
- This tutorial, designed for users of `Observable Javascript`, but offering many interesting examples for `Pandas` aficionados;
- The following cheatsheet from this post.

As a reminder, to execute the code examples in an interactive notebook, you can use the shortcuts at the top of the page to launch it in your preferred environment.

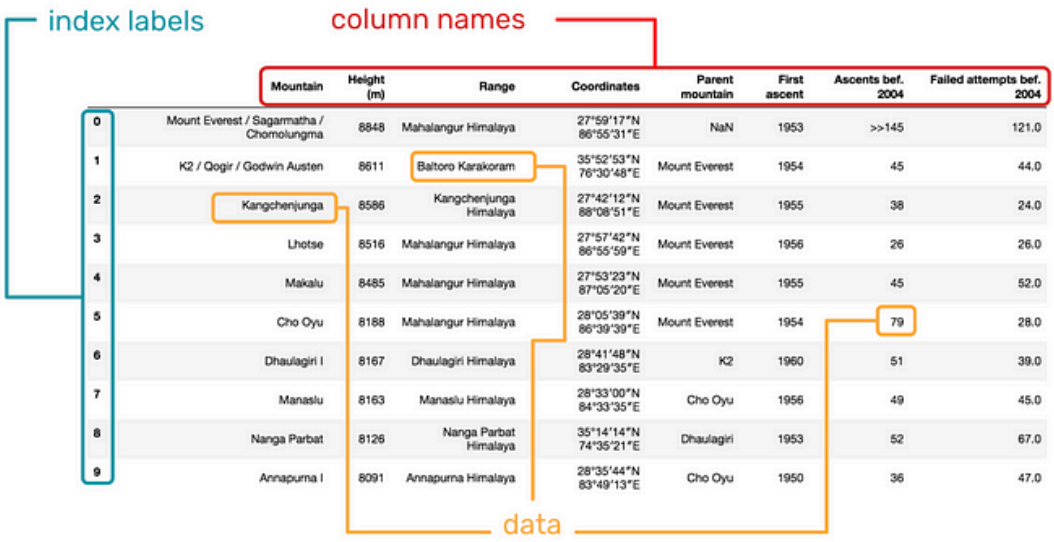
2 `Pandas` logic

2.1 Anatomy of a `Pandas` table

The central object in the `Pandas` logic is the `DataFrame`. It is a special data structure with two dimensions, structured by aligning rows and columns. Unlike a matrix, columns can be of different types.

A `DataFrame` consists of the following elements:

- the row index;
- the column name;
- the data value;



	Mountain	Height (m)	Range	Coordinates	Parent mountain	First ascent	Ascents bef. 2004	Failed attempts bef. 2004
0	Mount Everest / Sagarmatha / Chomolungma	8848	Mahalangur Himalaya	27°59'17"N 86°55'31"E	NaN	1953	>>145	121.0
1	K2 / Qogir / Godwin Austen	8611	Baltoro Karakoram	35°52'53"N 76°30'48"E	Mount Everest	1954	45	44.0
2	Kangchenjunga	8586	Kangchenjunga Himalaya	27°42'12"N 88°08'51"E	Mount Everest	1955	38	24.0
3	Lhotse	8516	Mahalangur Himalaya	27°57'42"N 86°55'59"E	Mount Everest	1956	26	26.0
4	Makalu	8485	Mahalangur Himalaya	27°53'23"N 87°05'20"E	Mount Everest	1955	45	52.0
5	Cho Oyu	8188	Mahalangur Himalaya	28°05'39"N 86°39'39"E	Mount Everest	1954	79	28.0
6	Dhaulagiri I	8167	Dhaulagiri Himalaya	28°41'48"N 83°29'35"E	K2	1960	51	39.0
7	Manaslu	8163	Manaslu Himalaya	28°33'00"N 84°33'35"E	Cho Oyu	1956	49	45.0
8	Nanga Parbat	8126	Nanga Parbat Himalaya	35°14'14"N 74°35'21"E	Dhaulagiri	1953	52	67.0
9	Annapurna I	8091	Annapurna Himalaya	28°35'44"N 83°49'13"E	Cho Oyu	1950	36	47.0

Figure 1: Structure of a `Pandas` *DataFrame*, borrowed from https://x.com/epfl_exts/status/997506000600084480

2.2 Before seeing `DataFrame`, we need to know `Pandas` `Series`

In fact, a *DataFrame* is a collection of objects called `pandas.Series`. These `Series` are one-dimensional

objects that are extensions of the one-dimensional `Numpy` arrays[3]. In particular, to facilitate the handling of categorical or temporal data, additional variable types are available in `Pandas` compared to `Numpy` (`categorical`, `datetime64`, and `timedelta64`). These types are associated with optimized methods to facilitate the processing of this data.

There are several possible types for a `pandas.Series`, extending the basic data types in `Python`, which will determine the behavior of this variable. Indeed, many operations do not have the same meaning depending on whether the value is numeric or not.

The simplest types (`int` or `float`) correspond to numeric values:

```
poids = pd.Series(
    [3, 7, 12]
)
poids
```

```
0     3
1     7
2    12
dtype: int64
```

> ! Missing Values are tricky !
>
> In general, if `Pandas` exclusively detects integer values in a variable, it will use the `int` type to optimize memory. This choice makes sense. However, it has a drawback: `Numpy`, and therefore by extension `Pandas`, cannot represent missing values for the `int` type (more on missing values below).
>
> Pending the shift to `Arrow`, which can handle missing values in `int`, the method to use is to convert to the `float` type if the variable will have missing values, which is quite simple:

For textual data, it is just as simple:

```
animal = pd.Series(
  ['chat', 'chien', 'koala']
)
animal
```

```
0     chat
1    chien
2    koala
dtype: str
```

Since `Pandas` 3.0, exclusively textual data natively receives the dedicated `str` type, as shown above: `object` is no longer the default type used for text. The `object` type remains a catch-all, but it is now reserved for genuinely mixed data (text and numbers, or arbitrary `Python` objects, type `mixed`). Historically, `object` acted as an intermediate type between the `factor` and `character` of `R`; this role is now better served by the `str`/`category` distinction. There is indeed an equivalent type to `factor` in `Pandas`, the `category` type, to be preferred when a variable has a finite and relatively small list of possible

[3] The original goal of `Pandas` is to provide a high-level library for more abstract low-level layers, such as `Numpy` arrays. `Pandas` is gradually changing these low-level layers to favor `Arrow` over `Numpy` without destabilizing the high-level commands familiar to `Pandas` users. This shift is due to the fact that `Arrow`, a low-level computation library, is more powerful and flexible than `Numpy`. For example, `Numpy` offers limited textual types, whereas `Arrow` provides greater freedom.

values (bringing memory and performance gains). It is therefore recommended, when focusing on a low-cardinality textual variable, to convert it explicitly:

```
animal.astype("category")
animal.astype(str)
```

Line 1
To convert to `category` (the choice that makes sense here)

Line 2
To explicitly convert back to `str` (redundant here since this is already the default type since `Pandas` 3.0, but useful if the column comes from an import that forced an `object` typing)

```
0     chat
1    chien
2    koala
dtype: str
```

It is important to examine the types of your `Pandas` objects and convert them if they do not make sense; `Pandas` makes optimized choices, but it may be necessary to correct them because `Pandas` does not know your future data usage. This is one of the tasks to do during *feature engineering*, the set of steps for preparing data for future use.

3 From `Series` to `DataFrame`

We have created two independent series, `animal` and `poids`, which are related. In the matrix world, this would correspond to moving from a vector to a matrix. In the `Pandas` world, this means moving from a `Series` to a `DataFrame`.

This is done naturally with `Pandas`:

```
animaux = pd.DataFrame(
  zip(animal, poids),
  columns = ['animal','poids']
)
animaux
```

	animal	poids
0	chat	3
1	chien	7
2	koala	12

1. We need to use `zip` here because `Pandas` expects a structure like `{"var1": [val1, val2], "var2": [val1, val2]}`, which is not what we prepared previously. However, we will see that this approach is not the most common for creating a `DataFrame`.

The essential difference between a `Series` and a `Numpy` object is indexing. In `Numpy`, indexing is implicit; it allows accessing data (the one at the index located at position *i*). With a `Series`, you can, of course, use a positional index, but more importantly, you can use more explicit indices.

This allows accessing data more naturally, using column names, for example:

```
animaux['poids']
```

```
0     3
1     7
2    12
Name: poids, dtype: int64
```

The existence of an index makes subsetting, that is, selecting rows or columns, particularly easy. DataFrames have two indices: those for rows and those for columns. You can make selections on both dimensions. Anticipating later exercises, you can see that this will facilitate row selection:

```
animaux.loc[animaux['animal'] == "chat", 'poids']
```

```
0    3
Name: poids, dtype: int64
```

This instruction is equivalent to the SQL command:

```
SELECT poids FROM animaux WHERE animal == "chat"
```

If we return to our `animaux` dataset, we can see the row number displayed on the left:

```
animaux
```

	animal	poids
0	chat	3
1	chien	7
2	koala	12

This is the default index for the row dimension because we did not configure one. It is not mandatory; it is quite possible to have an index corresponding to a variable of interest (we will discover this when we explore `groupby` in the next chapter). However, this can be tricky, and it is recommended that this be only transitory, hence the importance of regularly performing `reset_index`.

3.1 The concept of *tidy data*

The concept of ***tidy*** data, popularized by Hadley Wickham through his `R` packages (see H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund [2]), is highly relevant for describing the structure of a `Pandas` DataFrame. The three rules of tidy data are as follows:

- Each **variable** has its own column;
- Each **observation** has its own row;
- A **value**, representing an observation of a variable, is located in a single cell.

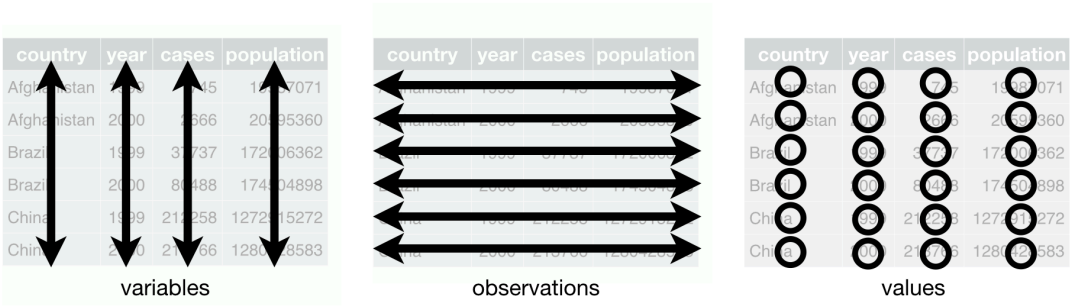


Figure 2: Illustration of the *tidy data* concept (borrowed from H. Wickham)

These principles may seem like common sense, but you will find that many data formats do not adhere to them. For example, Excel spreadsheets often have values spanning multiple columns or several merged rows. Restructuring this data according to the tidy data principle will be crucial to performing analysis on it.

4 Importing Data with `Pandas`

If you had to manually create all your DataFrames from vectors, `Pandas` would not be practical. `Pandas` offers many functions to read data stored in different formats.

The simplest data to read are tabular data stored in an appropriate format. The two main formats to know are `CSV` and `Parquet`. The former has the advantage of simplicity - it is universal, known to all, and readable by any text editor. The latter is gaining popularity in the data ecosystem because it addresses some limitations of CSV (optimized storage, predefined variable types...) but has the disadvantage of not being readable without a suitable tool (which fortunately are increasingly present in standard code editors like `VSCode`). For more information on the difference between `CSV` and `Parquet`, refer to the deep dive chapters on the subject.

Data stored in other text-derived formats from `CSV` (`.txt`, `.tsv`...) or formats like `JSON` are readable with `Pandas`, but sometimes it takes a bit of iteration to find the correct reading parameters. In other chapters, we will discover that other data formats related to different data structures are also readable with `Python`.

Flat formats (`.csv`, `.txt`...) and the `Parquet` format are easy to use with `Pandas` because they are not proprietary and they store data in a tidy form. Data from spreadsheets, `Excel` or `LibreOffice`, are more or less complicated to import depending on whether they follow this schema or not. This is because these tools are used indiscriminately. While in the data science world, they should primarily be used to disseminate final tables for reporting, they are often used to disseminate raw data that could be better shared through more suitable channels.

One of the main difficulties with spreadsheets is that the data are usually associated with documentation in the same tab - for example, the spreadsheet has some lines describing the sources before the table - which requires human intelligence to assist `Pandas` during the import phase. This will always be possible, but when there is an alternative in the form of a flat file, there should be no hesitation.

4.1 Reading Data from a Local Path

This exercise aims to demonstrate the benefit of using a relative path rather than an absolute path to enhance code reproducibility. However, we will later recommend reading directly from the internet when possible and when it does not involve the recurrent download of a large file.

To prepare for this exercise, the following code will allow you to download data and write it locally:

```
import requests

url = "https://www.insee.fr/fr/statistiques/fichier/6800675/v_commune_2023.csv"
url_backup = "https://minio.lab.sspcloud.fr/lgaliana/data/python-ENSAE/cog_2023.csv"

try:
    response = requests.get(url)
except requests.exceptions.RequestException as e:
    print(f"Error : {e}")
    response = requests.get(url_backup)

# Only download if one of the request succeeded
if response.status_code == 200:
    with open("cog_2023.csv", "wb") as file:
        file.write(response.content)
```

Preliminary Exercise: Importing a CSV (Optional)

1. Use the code above ☝️ to download the data. Use `Pandas` to read the downloaded file.
2. Find where the data was written. Observe the structure of this directory.
3. Create a new folder using the file explorer (on the left in `Jupyter` or `VSCode`). Move the CSV and the notebook. Restart the kernel and adjust your code if needed. Repeat this process several times with different folders. What issues might you encounter?

	TYPECOM	COM	REG	DEP	CTCD	ARR	TNCC	NCC	NCCENR	LIBELLE	CAN	COMPARENT
0	COM	01001	84.0	01	01D	012	5	ABERGEMENT CLEMENCIAT	Abergement-Clémenciat	L'Abergement-Clémenciat	0108	NaN
1	COM	01002	84.0	01	01D	011	5	ABERGEMENT DE VAREY	Abergement-de-Varey	L'Abergement-de-Varey	0101	NaN

The main issue with reading from files stored locally is the risk of becoming dependent on a file system that is not necessarily shared. It is better, whenever possible, to directly read the data with an `HTTPS` link, which `Pandas` can handle. Moreover, when working with open data, this ensures that you are using the latest available data and not a local duplication that may not be up to date.

4.2 Reading from a CSV Available on the Internet

The URL to access the data can be stored in an ad hoc variable:

```
url = "https://koumoul.com/s/data-fair/api/v1/datasets/igt-pouvoir-de-rechauffement-global/convert"
```

The goal of the next exercise is to get familiar with importing and displaying data using `Pandas` and displaying a few observations.

Exercise 1: Importing a CSV and Exploring Data Structure

1. Import the data from Ademe using the `Pandas` package and the dedicated command for importing CSVs. Name the obtained `DataFrame emissions`[4].

[4]Due to a lack of imagination, we are often tempted to call our main *dataframe* `df` or `data`. This is often a bad idea because the name is not very informative when you read the code a few weeks later. Self-documenting code, an approach that consists of having code that is self-explanatory, is a good practice, and it is recommended to give a simple yet effective name to know the nature of the dataset in question.

2. Use the appropriate methods to display the first 10 values, the last 15 values, and a random sample of 10 values using the appropriate methods from the `Pandas` package.
3. Draw 5 percent of the sample without replacement.
4. Keep only the first 10 rows and randomly draw from these to obtain a DataFrame of 100 data points.
5. Make 100 draws from the first 6 rows with a probability of 1/2 for the first observation and a uniform probability for the others.

If you get stuck on question 1 Read the documentation for `read_csv` (very well done) or look for examples online to discover this function.

As illustrated by this exercise, displaying DataFrames in notebooks is quite ergonomic. The first and last rows are displayed automatically. For valuation tables present in a report or research article, the next chapter introduces `great_tables`, which offers very rich table formatting features.

⚠ Warning

Be careful with `display` and commands that reveal data (`head`, `tail`, etc.) in a notebook that handles confidential data when using version control software like `Git` (see dedicated chapters).

Indeed, you may end up sharing data inadvertently in the `Git` history. As explained in the chapter dedicated to `Git`, a file named `.gitignore` is sufficient to create some rules to avoid unintentional sharing of data with `Git`.

5 Exploring the Structure of a DataFrame

`Pandas` offers a data schema quite familiar to users of statistical software like `R`. Similar to the main data processing paradigms like the *tidyverse* (`R`), the grammar of `Pandas` inherits from `SQL` logic. The philosophy is very similar: operations are performed to select rows, columns, sort rows based on column values, apply standardized treatments to variables, etc. Generally, operations that reference variable names are preferred over those that reference row or column numbers.

Whether you are familiar with `SQL` or `R`, you will find a similar logic to what you know, although the names might differ: `df.loc[df['y']=='b']` may be written as `df %>% filter(y=='b')` (`R`) or `SELECT * FROM df WHERE y == 'b'` (`SQL`), but the logic is the same.

`Pandas` offers a plethora of pre-implemented functionalities. It is highly recommended, before writing a function, to consider if it is natively implemented in `Numpy`, `Pandas`, etc. Most of the time, if a solution is implemented in a library, it should be used as it will be more efficient than what you would implement.

To present the most practical methods for data analysis, we can use the example of the municipal CO2 consumption data from Ademe, which was the focus of the previous exercises.

```
df = pd.read_csv("https://koumoul.com/s/data-fair/api/v1/datasets/igt-pouvoir-de-rechauffement-global/convert")
df
```

	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire
0	01001	L'ABERGEMENT-CLEMENCIAT	3711.425991	NaN	NaN	432.751835	101.430476	2.354558	6.911213	309.358195	793.156501	367.036172
1	01002	L'ABERGEMENT-DE-VAREY	475.330205	NaN	NaN	140.741660	140.675439	2.354558	6.911213	104.866444	348.997893	112.934207
2	01004	AMBERIEU-EN-BUGEY	499.043526	212.577908	NaN	10313.446515	5314.314445	998.332482	2930.354461	16616.822534	15642.420313	10732.376934
3	01005	AMBERIEUX-EN-DOMBES	1859.160954	NaN	NaN	1144.429311	216.217508	94.182310	276.448534	663.683146	1756.341319	782.404357
4	01006	AMBLEON	448.966808	NaN	NaN	77.033834	48.401549	NaN	NaN	43.714019	398.786800	51.681756
...	...	...	...	...	...	...	...	...	...	...	...	...
35793	95676	VILLERS-EN-ARTHIES	1628.065094	NaN	NaN	165.045396	65.063617	11.772789	34.556067	176.098160	309.627908	235.439109
35794	95678	VILLIERS-ADAM	698.630772	NaN	NaN	1331.126598	111.480954	2.354558	6.911213	1395.529811	18759.370071	403.404815
35795	95680	VILLIERS-LE-BEL	107.564967	NaN	NaN	8367.174532	225.622903	534.484607	1568.845431	22613.830247	12217.122402	13849.512001
35796	95682	VILLIERS-LE-SEC	1090.890170	NaN	NaN	326.748418	108.969749	2.354558	6.911213	67.235487	4663.232127	85.657725
35797	95690	WY-DIT-JOLI-VILLAGE	1495.103542	NaN	NaN	125.236417	97.728612	4.709115	13.822427	117.450851	504.400972	147.867245

5.1 Dimensions and Structure of a DataFrame

The first useful methods allow displaying some attributes of a `DataFrame`.

```
df.axes
```

```
[RangeIndex(start=0, stop=35798, step=1),
 Index(['INSEE commune', 'Commune', 'Agriculture', 'Autres transports',
        'Autres transports international', 'CO2 biomasse hors-total', 'Déchets',
        'Energie', 'Industrie hors-énergie', 'Résidentiel', 'Routier',
        'Tertiaire'],
       dtype='str')]
```

```
df.columns
```

```
Index(['INSEE commune', 'Commune', 'Agriculture', 'Autres transports',
       'Autres transports international', 'CO2 biomasse hors-total', 'Déchets',
       'Energie', 'Industrie hors-énergie', 'Résidentiel', 'Routier',
       'Tertiaire'],
      dtype='str')
```

```
df.index
```

```
RangeIndex(start=0, stop=35798, step=1)
```

5.2 Dimensions and Structure of a DataFrame

The first useful methods allow displaying some attributes of a `DataFrame`.

```
df.ndim
```

```
2
```

```
df.shape
```

```
(35798, 12)
```

```
df.size
```

```
429576
```

To know the dimensions of a DataFrame, some practical methods can be used:

```
df.ndim
```

```
2
```

```
df.shape
```

```
(35798, 12)
```

```
df.size
```

```
429576
```

To determine the number of unique values of a variable, rather than writing a function yourself, use the `nunique` method. For example,

```
df['Commune'].nunique()
```

```
33338
```

`Pandas` offers many useful methods. Here is a summary of those related to data structure, accompanied by a comparison with `R`:

Operation	pandas	dplyr (`R`)
Retrieve column names	`df.columns`	`colnames(df)`
Retrieve dimensions	`df.shape`	`dim(df)`
Retrieve unique values of a variable	`df['myvar'].nunique()`	`df %>% summarise(distinct(myvar))`

5.3 Accessing elements of a DataFrame

In SQL, performing operations on columns is done with the `SELECT` command. With `Pandas`, to access an entire column, several approaches can be used:

- `dataframe.variable`, for example `df.Energie`. This method requires column names without spaces or special characters, which excludes many real datasets. It is not recommended.
- `dataframe[['variable']]` to return the variable as a `DataFrame`. This method can be tricky for a single variable; it's better to use `dataframe.loc[:,['variable']]`, which is more explicit about the nature of the resulting object.
- `dataframe['variable']` to return the variable as a `Series`. For example, `df[['Autres transports']]` or `df['Autres transports']`. This is the preferred method.

To retrieve multiple columns at once, there are two approaches, with the second being preferable:

- `dataframe[['variable1', 'variable2']]`
- `dataframe.loc[:, ['variable1', 'variable2']]`

This is equivalent to `SELECT variable1, variable2 FROM dataframe` in SQL.

Using `.loc` may seem excessively verbose, but it ensures that you are performing a subset operation on the column dimension. DataFrames have two indices, for rows and columns, and implicit operations can sometimes cause surprises, so it's more reliable to be explicit.

5.4 Accessing rows

To access one or more values in a `DataFrame`, there are two recommended methods, depending on the form of the row or column indices used:

- `df.iloc`: uses indices. This method is somewhat unreliable because a DataFrame's indices can change during processing (especially when performing group operations).
- `df.loc`: uses labels. This method is recommended.

> ⚠ Warning
>
> Code snippets using the `df.ix` structure should be avoided as the function is deprecated and may disappear at any time.

`iloc` refers to the indexing from 0 to *N*, where *N* equals `df.shape[0]` of a `pandas.DataFrame`. `loc` refers to the values of `df`'s index. For example, with the `pandas.DataFrame` `df_example`:

```
df_example = pd.DataFrame(
    {'month': [1, 4, 7, 10], 'year': [2012, 2014, 2013, 2014], 'sale': [55, 40, 84, 31]})
df_example = df_example.set_index('month')
df_example
```

	year	sale
month		
1	2012	55
4	2014	40
7	2013	84
10	2014	31

- `df_example.loc[1, :]` will return the first row of `df` (row where the `month` index equals 1);
- `df_example.iloc[1, :]` will return the second row (since Python indexing starts at 0);
- `df_example.iloc[:, 1]` will return the second column, following the same principle.

Later exercises will allow practicing this syntax on our dataset of carbon emissions.

6 Main data wrangling routine

The most frequent operations in `SQL` are summarized in the following table. It is useful to know them (many data manipulation syntaxes use these terms) because, one way or another, they cover most data manipulation uses. We will describe some of them later:

Operation	SQL	pandas	dplyr (`R`)
Select variables by name	`SELECT`	`df.loc[:, ['Autres trans`	`df %>% select(Autres transports, Energie)`
Select observations based on one or more conditions	`FILTER`	`df.loc[pd.col('Agricult`	`df %>% filter(Agriculture>2000)`

Operation	SQL	pandas	dplyr (R)
Sort the table by one or more variables	`SORT BY`	`df.sort_values(['Commun`	`df %>% arrange(Commune, Agriculture)`
Add variables that are functions of other variables	`SELECT *, LOG(A AS x FROM df`	`df = df.assign(x = np.lo`	`df %>% mutate(x = log(Agriculture))`

The next chapters will let us go further by adding other, more complex operations (joins, group-by statistics, etc.)

6.1 Operations on Columns: Adding or Removing Variables, Renaming Them, etc.

Technically, `Pandas` DataFrames are *mutable* objects in the `Python` language, meaning it is possible to change the DataFrame as needed during processing.

The most classic operation is adding or removing variables to the data table. The simplest way to add columns is by reassignment. For example, to create a `dep` variable that corresponds to the first two digits of the commune code (INSEE code), simply take the variable and apply the appropriate treatment (in this case, keep only its first two characters):

```
df['dep'] = df['INSEE commune'].str[:2]
df.head(3)
```

	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep
0	01001	L'ABERGEMENT-CLEMENCIAT	3711.425991	NaN	NaN	432.751835	101.430476	2.354558	6.911213	309.358195	793.156501	367.036172	01
1	01002	L'ABERGEMENT-DE-VAREY	475.330205	NaN	NaN	140.741660	140.675439	2.354558	6.911213	104.866444	348.997893	112.934207	01
2	01004	AMBERIEU-EN-BUGEY	499.043526	212.577908	NaN	10313.446515	5314.314445	998.332482	2930.354461	16616.822534	15642.420313	10732.376934	01

In SQL, the method depends on the execution engine. In pseudo-code, it would be:

```
SELECT everything(), SUBSTR("code_insee", 2) AS dep FROM df
```

It is possible to apply this approach to creating columns on multiple columns. One of the advantages of this approach is that it allows recycling column names.

```
vars = ['Agriculture', 'Déchets', 'Energie']

df[[v + "_log" for v in vars]] = np.log(df.loc[:, vars])
df.head(3)
```

	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep	Agriculture_log	Déchets_log	Energie_log
0	01001	L'ABERGEMENT-CLEMENCIAT	3711.425991	NaN	NaN	432.751835	101.430476	2.354558	6.911213	309.358195	793.156501	367.036172	01	8.219171	4.619374	0.856353
1	01002	L'ABERGEMENT-DE-VAREY	475.330205	NaN	NaN	140.741660	140.675439	2.354558	6.911213	104.866444	348.997893	112.934207	01	6.164010	4.946455	0.856353
2	01004	AMBERIEU-EN-BUGEY	499.043526	212.577908	NaN	10313.446515	5314.314445	998.332482	2930.354461	16616.822534	15642.420313	10732.376934	01	6.212693	8.578159	6.906086

The equivalent SQL query would be quite tedious to write. For such operations, the benefit of a high-level library like `Pandas` becomes clear.

> ⚠ Warning
>
> This is possible thanks to the native vectorization of `Numpy` operations and the magic of `Pandas` that rearranges everything. This is not usable with just any function. For other functions, you will need to use `assign`, generally through lambda functions, temporary functions acting as pass-throughs. For example, to create a variable using this approach, you would do:

```
df.assign(
  Energie_log = lambda x: np.log(x['Energie'])
)
```

Since `Pandas` 3.0, `pd.col` lets you avoid the lambda function while still using `assign`:

```
df.assign(
  Energie_log = np.log(pd.col('Energie'))
)
```

With methods from `Pandas` or `Numpy` like this, it is not beneficial and even counterproductive as it slows down the code.

Variables can be easily renamed using the `rename` method, which works well with dictionaries. To rename columns, specify the parameter `axis = 'columns'` or `axis=1`. The `axis` parameter is often necessary because many `Pandas` methods assume by default that operations are done on the row index:

```
df = df.rename({"Energie": "eneg", "Agriculture": "agr"}, axis=1)
df.head()
```

	INSEE commune	Commune	agr	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	eneg	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep	Agriculture_log	Déchets_log	Energie_log
0	01001	L'ABERGEMENT-CLEMENCIAT	3711.425991	NaN	NaN	432.751835	101.430476	2.354558	6.911213	309.358195	793.156501	367.036172	01	8.219171	4.619374	0.856353
1	01002	L'ABERGEMENT-DE-VAREY	475.330205	NaN	NaN	140.741660	140.675439	2.354558	6.911213	104.866444	348.997893	112.934207	01	6.164010	4.946455	0.856353
2	01004	AMBERIEU-EN-BUGEY	499.043526	212.577908	NaN	10313.446515	5314.314445	998.332482	2930.354461	16616.822534	15642.420313	10732.376934	01	6.212693	8.578159	6.906086
3	01005	AMBERIEUX-EN-DOMBES	1859.160954	NaN	NaN	1144.429311	216.217508	94.182310	276.448534	663.683146	1756.341319	782.404357	01	7.527881	5.376285	4.545232
4	01006	AMBLEON	448.966808	NaN	NaN	77.033834	48.401549	NaN	NaN	43.714019	398.786800	51.681756	01	6.106949	3.879532	NaN

Finally, to delete columns, use the `drop` method with the `columns` argument:

```
df = df.drop(columns = ["eneg", "agr"])
```

6.2 Reordering observations

The `sort_values` method allows reordering observations in a `DataFrame`, keeping the column order the same.

For example, to sort in descending order of CO2 consumption in the residential sector, you would do:

```
df = df.sort_values("Résidentiel", ascending = False)
df.head(3)
```

	INSEE commune	Commune	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep	Agriculture_log	Déchets_log	Energie_log
12167	31555	TOULOUSE	4482.980062	130.792683	576394.181208	88863.732538	277062.573234	410675.902028	586054.672836	288175.400126	31	7.268255	11.394859	11.424640
16774	44109	NANTES	138738.544337	250814.701179	193478.248177	18162.261628	77897.138554	354259.013785	221068.632724	173447.582779	44	5.513507	9.807101	9.767748
27294	67482	STRASBOURG	124998.576639	122266.944279	253079.442156	119203.251573	135685.440035	353586.424577	279544.852332	179562.761386	67	6.641974	11.688585	9.885411

Thus, in one line of code, you can identify the cities where the residential sector consumes the most. In SQL, you would do:

```
SELECT * FROM df ORDER BY "Résidentiel" DESC
```

6.3 Filtering

The operation of selecting rows is called `FILTER` in SQL. It is used based on a logical condition (clause `WHERE`). Data is selected based on a logical condition.

There are several methods in `Pandas`. The simplest is to use boolean masks, as seen in the chapter `numpy`.

For example, to select the municipalities in Hauts-de-Seine, you can start by using the result of the `str.startswith` method (which returns `True` or `False`):

```
df['INSEE commune'].str.startswith("92")
```

```
12167    False
16774    False
27294    False
12729    False
22834    False
         ...
20742    False
20817    False
20861    False
20898    False
20957    False
Name: INSEE commune, Length: 35798, dtype: bool
```

`str.` is a special method in `Pandas` that allows treating each value of a vector as a native `string` in `Python` to which a subsequent method (in this case, `startswith`) is applied.

The above instruction returns a vector of booleans. We previously saw that the `loc` method is used for subsetting on both row and column indices. It works with boolean vectors. In this case, if subsetting on the row dimension (or column), it will return all observations (or variables) that satisfy this condition.

Thus, by combining these two elements, we can filter our data to get only the results for the 92:

```
df.loc[df['INSEE commune'].str.startswith("92")].head(2)
```

	INSEE commune	Commune	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep	Agriculture_log	Déchets_log	Energie_log
35494	92012	BOULOGNE-BILLANCOURT	1250.483441	34.234669	51730.704250	964.828694	25882.493998	92216.971456	64985.280901	60349.109482	92	NaN	6.871951	9.084530
35501	92025	COLOMBES	411.371588	14.220061	53923.847088	698.685861	50244.664227	87469.549463	52070.927943	41526.600867	92	NaN	6.549201	9.461557

Since `Pandas` 3.0, an alternative notation exists, `pd.col`, which lets you reference a column without repeating the `DataFrame` name. It should be preferred as it makes code more readable and easier to chain (it is close to `Polars`' `pl.col` notation):

```
df.loc[pd.col('INSEE commune').str.startswith("92")].head(2)
```

	INSEE commune	Commune	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep	Agriculture_log	Déchets_log	Energie_log
35494	92012	BOULOGNE-BILLANCOURT	1250.483441	34.234669	51730.704250	964.828694	25882.493998	92216.971456	64985.280901	60349.109482	92	NaN	6.871951	9.084530
35501	92025	COLOMBES	411.371588	14.220061	53923.847088	698.685861	50244.664227	87469.549463	52070.927943	41526.600867	92	NaN	6.549201	9.461557

The equivalent SQL code may vary depending on the execution engine (`DuckDB`, `PostGre`, `MySQL`) but would take a form similar to this:

```
SELECT * FROM df WHERE STARTSWITH("INSEE commune", "92")
```

6.4 Summary of main operations

The main manipulations are as follows:

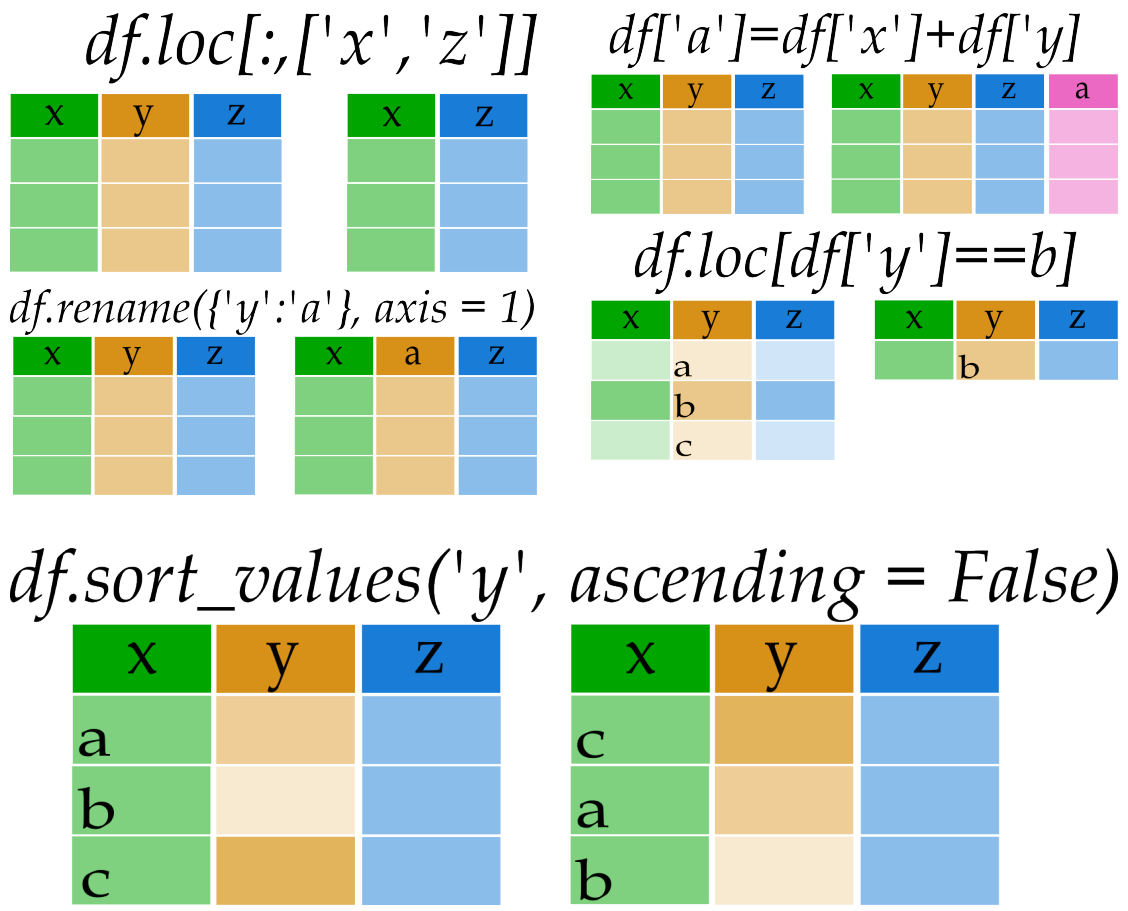


Figure 1: Reordering the DataFrame

7 Descriptive Statistics

To start again from the raw source, let's recreate our dataset for the examples:

```
df = pd.read_csv("https://koumoul.com/s/data-fair/api/v1/datasets/igt-pouvoir-de-rechauffement-global/convert")
df.head(3)
```

	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire
0	01001	L'ABERGEMENT-CLEMENCIAT	3711.425991	NaN	NaN	432.751835	101.430476	2.354558	6.911213	309.358195	793.156501	367.036172
1	01002	L'ABERGEMENT-DE-VAREY	475.330205	NaN	NaN	140.741660	140.675439	2.354558	6.911213	104.866444	348.997893	112.934207
2	01004	AMBERIEU-EN-BUGEY	499.043526	212.577908	NaN	10313.446515	5314.314445	998.332482	2930.354461	16616.822534	15642.420313	10732.376934

`Pandas` includes several methods to construct aggregate statistics: sum, count of unique values, count of non-missing values, mean, variance, etc.

The most generic method is `describe`

```
df.describe()
```

	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire
count	35736.000000	9979.000000	2.891000e+03	35798.000000	35792.000000	3.449000e+04	3.449000e+04	35792.000000	35778.000000	35798.000000
mean	2459.975760	654.919940	7.692345e+03	1774.381550	410.806329	6.625698e+02	2.423128e+03	1783.677872	3535.501245	1105.165915
std	2926.957701	9232.816833	1.137643e+05	7871.341922	4122.472608	2.645571e+04	5.670374e+04	8915.902379	9663.156628	5164.182507
min	0.003432	0.000204	3.972950e-04	3.758088	0.132243	2.354558e+00	1.052998e+00	1.027266	0.555092	0.000000
25%	797.682631	52.560412	1.005097e+01	197.951108	25.655166	2.354558e+00	6.911213e+00	96.052911	419.700460	94.749885
50%	1559.381285	106.795928	1.992434e+01	424.849988	54.748653	4.709115e+00	1.382243e+01	227.091193	1070.895593	216.297718
75%	3007.883903	237.341501	3.298311e+01	1094.749825	110.820941	5.180027e+01	1.520467e+02	749.469293	3098.612157	576.155869
max	98949.317760	513140.971691	3.303394e+06	576394.181208	275500.374439	2.535858e+06	6.765119e+06	410675.902028	586054.672836	288175.400126

which is similar to the eponymous method in `Stata` or the `PROC FREQ` of `SAS`, two proprietary languages. However, it provides a lot of information, which can often be overwhelming. Therefore, it is more practical to work directly on a few columns or choose the statistics you want to use.

7.1 Counts

The first type of statistics you might want to implement involves counting or enumerating values.

For example, if you want to know the number of municipalities in your dataset, you can use the `count` method or `nunique` if you are interested in unique values.

```
df['Commune'].count()
```

```
np.int64(35798)
```

```
df['Commune'].nunique()
```

```
33338
```

In SQL, the first instruction would be `SELECT COUNT(Commune) FROM df`, the second `SELECT COUNT DISTINCT Commune FROM df`. Here, this allows us to understand that there may be duplicates in the `Commune` column, which needs to be considered if we want to uniquely identify our municipalities (this will be addressed in the next chapter on data merging).

The coherence of the `Pandas` syntax allows you to do this for several columns simultaneously. In SQL, this would be possible but the code would start to become quite verbose:

```
df.loc[:, ['Commune', 'INSEE commune']].count()
```

```
Commune          35798
INSEE commune    35798
dtype: int64
```

```
df.loc[:, ['Commune', 'INSEE commune']].nunique()
```

```
Commune          33338
INSEE commune    35798
dtype: int64
```

With these two simple commands, we understand that our `INSEE commune` variable (the INSEE code) will be more reliable for identifying municipalities than the names, which are not necessarily unique. The purpose of the INSEE code is to provide a unique identifier, unlike the postal code which can be shared by several municipalities.

7.2 Aggregated statistics

`Pandas` includes several methods to construct statistics on multiple columns: sum, mean, variance, etc.

The methods are quite straightforward:

```
df['Agriculture'].sum()
df['Agriculture'].mean()
```

```
np.float64(2459.975759687974)
```

Again, the consistency of `Pandas` allows generalizing the calculation of statistics to multiple columns:

```
df.loc[:, ['Agriculture', 'Résidentiel']].sum()
df.loc[:, ['Agriculture', 'Résidentiel']].mean()
```

```
Agriculture    2459.975760
Résidentiel    1783.677872
dtype: float64
```

It is possible to generalize this to all columns. However, it is necessary to introduce the `numeric_only` parameter to perform the aggregation task only on the relevant variables:

```
df.mean(numeric_only = True)
```

```
Agriculture                        2459.975760
Autres transports                   654.919940
Autres transports international    7692.344960
CO2 biomasse hors-total            1774.381550
Déchets                             410.806329
Energie                             662.569846
Industrie hors-énergie             2423.127789
Résidentiel                        1783.677872
Routier                            3535.501245
Tertiaire                          1105.165915
dtype: float64
```

> ⚠ Warning
>
> Version 2.0 of `Pandas` introduced a change in the behavior of aggregation methods.
>
> It is now necessary to specify whether you want to perform operations exclusively on numeric columns. This is why we explicitly state the argument `numeric_only = True` here. This behavior was previously implicit.

The practical method to know is `agg`. This allows defining the statistics you want to calculate for each variable:

```
df.agg(
  {
    'Agriculture': ['sum', 'mean'],
    'Résidentiel': ['mean', 'std'],
    'Commune': 'nunique'
  }
)
```

	Agriculture	Résidentiel	Commune
sum	8.790969e+07	NaN	NaN
mean	2.459976e+03	1783.677872	NaN
std	NaN	8915.902379	NaN
nunique	NaN	NaN	33338.0

The output of aggregation methods is an indexed `Series` (methods like `df.sum()`) or directly a `DataFrame` (the `agg` method). It is generally more practical to have a `DataFrame` than an indexed `Series` if you want to rework the table to draw conclusions. Therefore, it is useful to transform the outputs from `Series` to `DataFrame` and then apply the `reset_index` method to convert the index into a column. From there, you can modify the `DataFrame` to make it more readable.

For example, if you are interested in the share of each sector in total emissions, you can proceed in two steps. First, create an observation per sector representing its total emissions:

```
# Etape 1: création d'un DataFrame propre
emissions_totales = (
  pd.DataFrame(
    df.sum(numeric_only = True),
    columns = ["emissions"]
  )
  .reset_index(names = "secteur")
)
emissions_totales
```

	secteur	emissions
0	Agriculture	8.790969e+07
1	Autres transports	6.535446e+06
2	Autres transports international	2.223857e+07
3	CO2 biomasse hors-total	6.351931e+07
4	Déchets	1.470358e+07
5	Energie	2.285203e+07
6	Industrie hors-énergie	8.357368e+07
7	Résidentiel	6.384140e+07
8	Routier	1.264932e+08
9	Tertiaire	3.956273e+07

Then, work minimally on the dataset to get some interesting conclusions about the structure of emissions in France:

```
emissions_totales['emissions (%)'] = (
  100*emissions_totales['emissions']/emissions_totales['emissions'].sum()
)
(emissions_totales
  .sort_values("emissions", ascending = False).
  round()
)
```

	secteur	emissions	emissions (%)
8	Routier	126493164.0	24.0
0	Agriculture	87909694.0	17.0

	secteur	emissions	emissions (%)
6	Industrie hors-énergie	83573677.0	16.0
7	Résidentiel	63841398.0	12.0
3	CO2 biomasse hors-total	63519311.0	12.0
9	Tertiaire	39562729.0	7.0
5	Energie	22852034.0	4.0
2	Autres transports international	22238569.0	4.0
4	Déchets	14703580.0	3.0
1	Autres transports	6535446.0	1.0

This table is not well-formatted and far from being presentable, but it is already useful from an exploratory perspective. It helps us understand the most emitting sectors, namely transport, agriculture, and industry, excluding energy. The fact that energy is relatively low in emissions can be explained by the French energy mix, where nuclear power represents a majority of electricity production.

To go further in formatting this table to have communicable statistics outside of `Python`, we will explore `great_tables` in the next chapter.

> **Note**
>
> The data structure resulting from `df.sum` is quite practical (it is tidy). We could do exactly the same operation as `df.sum(numeric_only = True)` with the following code:

```
df.select_dtypes(include='number').agg(func=sum)
```

```
Agriculture                          NaN
Autres transports                    NaN
Autres transports international      NaN
CO2 biomasse hors-total        6.351931e+07
Déchets                              NaN
Energie                              NaN
Industrie hors-énergie               NaN
Résidentiel                          NaN
Routier                              NaN
Tertiaire                      3.956273e+07
dtype: float64
```

7.3 Missing values

So far, we haven't discussed what could be a stumbling block for a data scientist: missing values.

Real datasets are rarely complete, and missing values can reflect many realities: data retrieval issues, irrelevant variables for a given observation, etc.

Technically, `Pandas` handles missing values without issues (except for `int` variables, but that's an exception). By default, missing values are displayed as `NaN` and are of type `np.nan` (for temporal values, i.e., of type `datetime64`, missing values are `NaT`). We get consistent aggregation behavior when combining two columns, one of which has missing values.

```
import numpy as np
rng = np.random.default_rng()
```

```
ventes = pd.DataFrame(
    {'prix': rng.uniform(size = 5),
     'client1': [i+1 for i in range(5)],
     'client2': [i+1 for i in range(4)] + [np.nan],
     'produit': [np.nan] + ['yaourt','pates','riz','tomates']
    }
)
ventes
```

	prix	client1	client2	produit
0	0.364815	1	1.0	NaN
1	0.101638	2	2.0	yaourt
2	0.770709	3	3.0	pates
3	0.900129	4	4.0	riz
4	0.951826	5	NaN	tomates

`Pandas` will refuse to aggregate because, for it, a missing value is not zero:

```
ventes["client1"] + ventes["client2"]
```

```
0    2.0
1    4.0
2    6.0
3    8.0
4    NaN
dtype: float64
```

It is possible to remove missing values using `dropna()`. This method will remove all rows where there is at least one missing value.

```
ventes.dropna()
```

	prix	client1	client2	produit
1	0.101638	2	2.0	yaourt
2	0.770709	3	3.0	pates
3	0.900129	4	4.0	riz

In this case, we lose two rows. It is also possible to remove only the columns where there are missing values in a DataFrame with `dropna()` using the `subset` parameter.

```
ventes.dropna(subset=["produit"])
```

	prix	client1	client2	produit
1	0.101638	2	2.0	yaourt
2	0.770709	3	3.0	pates
3	0.900129	4	4.0	riz
4	0.951826	5	NaN	tomates

This time we lose only one row, the one where `produit` is missing.

`Pandas` provides the ability to impute missing values using the `fillna()` method. For example, if you think the missing values in `produit` are zeros, you can do:

```
ventes.dropna(subset=["produit"]).fillna(0)
```

	prix	client1	client2	produit
1	0.101638	2	2.0	yaourt
2	0.770709	3	3.0	pates
3	0.900129	4	4.0	riz
4	0.951826	5	0.0	tomates

If you want to impute the median for the `client2` variable, you can slightly change this code by encapsulating the median calculation inside:

```
(ventes["client2"]
  .fillna(
    ventes["client2"].median()
  )
)
```

```
0    1.0
1    2.0
2    3.0
3    4.0
4    2.5
Name: client2, dtype: float64
```

For real-world datasets, it is useful to use the `isna` (or `isnull`) method combined with `sum` or `mean` to understand the extent of missing values in a dataset.

```
df.isnull().mean().sort_values(ascending = False)
```

```
Autres transports international    0.919241
Autres transports                  0.721241
Energie                            0.036538
Industrie hors-énergie             0.036538
Agriculture                        0.001732
Routier                            0.000559
Résidentiel                        0.000168
Déchets                            0.000168
INSEE commune                      0.000000
Commune                            0.000000
CO2 biomasse hors-total            0.000000
Tertiaire                          0.000000
dtype: float64
```

This preparatory step is useful for anticipating the question of imputation or filtering on missing values: are they missing at random or do they reflect an issue in data retrieval? The choices related to handling missing values are not neutral methodological choices. `Pandas` provides the technical tools to do this, but the legitimacy and relevance of these choices are specific to each dataset. Data explorations aim to detect clues to make an informed decision.

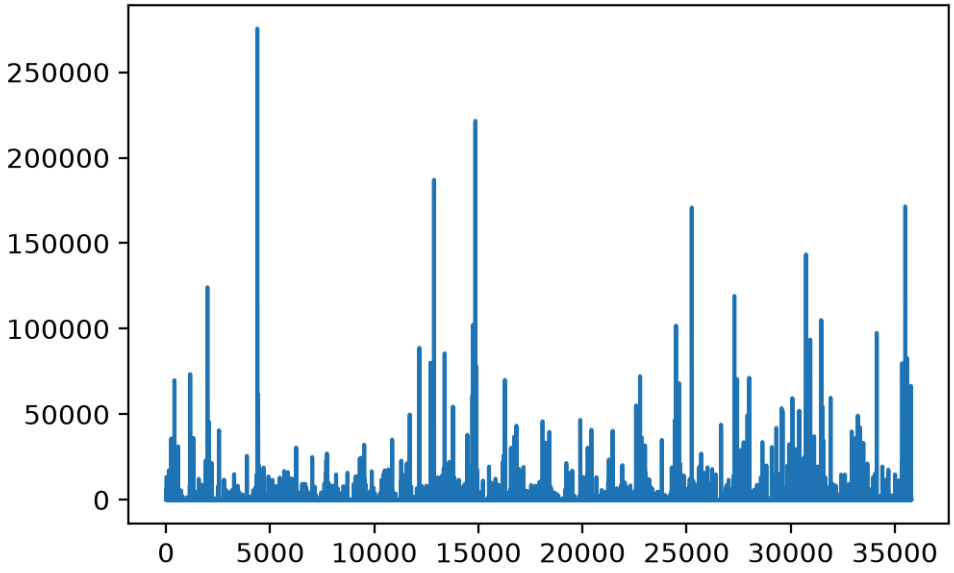
8 Quick graphical representations

Numerical tables are certainly useful for understanding the structure of a dataset, but their dense aspect makes them difficult to grasp. Having a simple graph can be useful to visualize the distribution of the data at a glance and thus understand the normality of an observation.

`Pandas` includes basic graphical methods to meet this need. They are practical for quickly producing a graph, especially after complex data manipulation operations. We will delve deeper into the issue of data visualizations in the Communicate section.

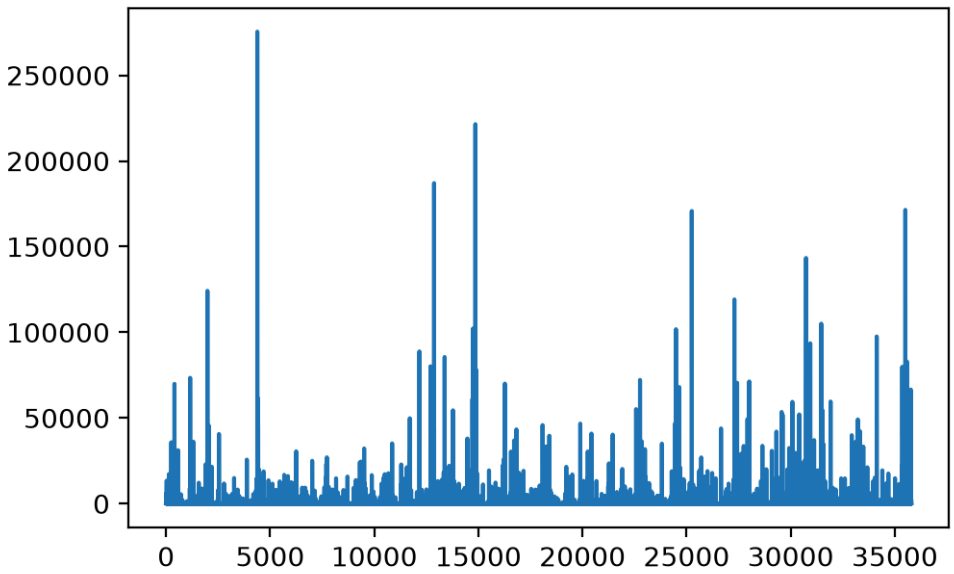
You can apply the `plot()` method directly to a `Series`:

```
df['Déchets'].plot()
```



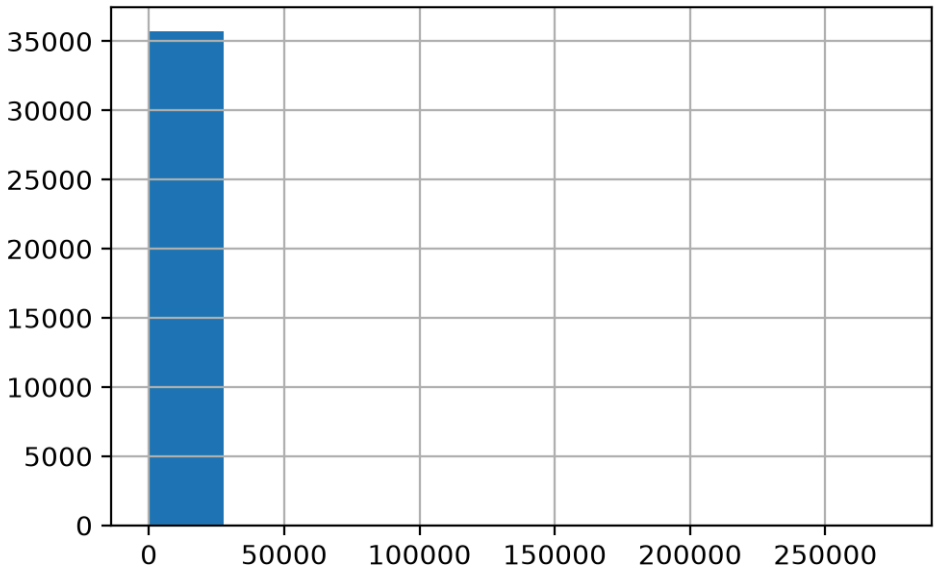
The equivalent code with `matplotlib` would be:

```
import matplotlib.pyplot as plt
plt.plot(df.index, df['Déchets'])
```



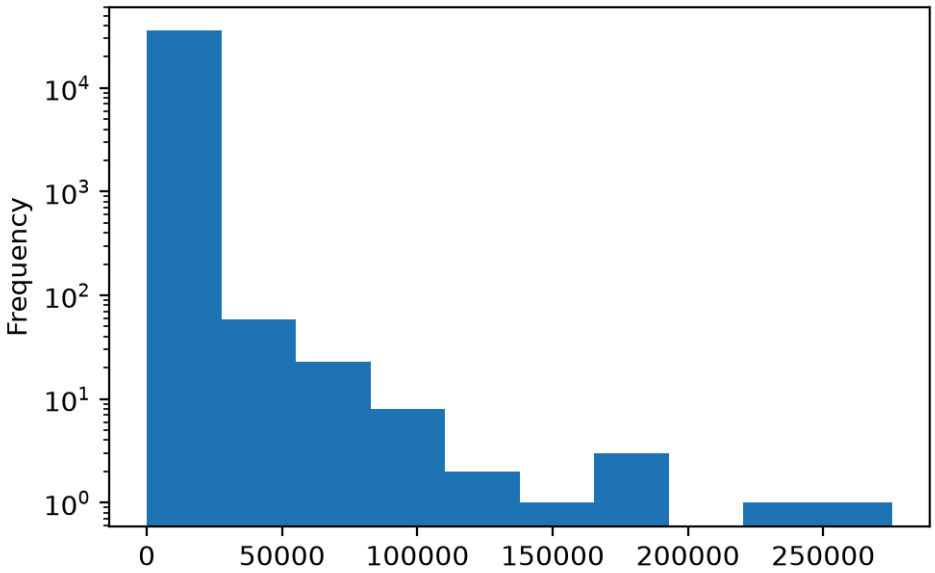
By default, the obtained visualization is a series. This is not necessarily what is expected since it only makes sense for time series. As a data scientist working with microdata, you are more often interested in a histogram to get an idea of the data distribution. To do this, simply add the argument `kind = 'hist'`:

```
df['Déchets'].hist()
```



With data that has a non-normalized distribution, which represents many real-world variables, histograms are generally not very informative. The log can be a solution to bring some extreme values to a comparable scale:

```
df['Déchets'].plot(kind = 'hist', logy = True)
```



The output is a `matplotlib` object. Customizing these figures is thus possible (and even desirable because the default `matplotlib` graphs are quite basic). However, this is a quick method for constructing figures that require work for a finalized visualization. This involves thorough work on the `matplotlib` object or using a higher-level library for graphical representation (`seaborn`, `plotnine`, `plotly`, etc.).

The part of this course dedicated to data visualization will briefly present these different visualization paradigms. These do not exempt you from using common sense in choosing the graph used to represent a descriptive statistic (see this conference by Eric Mauvière).

9 Synthesis exercise

This exercise synthesizes several steps of data preparation and exploration to better understand the structure of the phenomenon we want to study, namely carbon emissions in France.

It is recommended to start from a clean session (in a notebook, you should do `Restart Kernel`) to avoid an environment polluted by other objects. You can then run the following code to get the necessary base:

```
import pandas as pd

emissions = pd.read_csv("https://koumoul.com/s/data-fair/api/v1/datasets/igt-pouvoir-de-rechauffement-global/convert")
emissions.head(2)
```

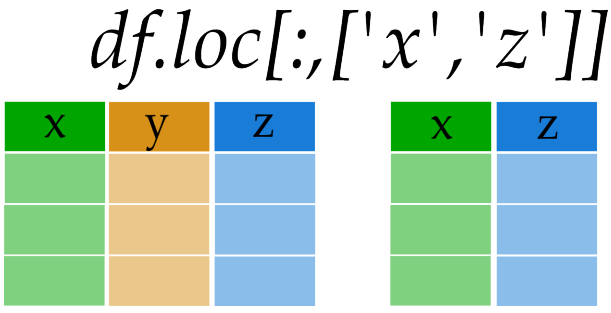
	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire
0	01001	L'ABERGEMENT-CLEMENCIAT	3711.425991	NaN	NaN	432.751835	101.430476	2.354558	6.911213	309.358195	793.156501	367.036172
1	01002	L'ABERGEMENT-DE-VAREY	475.330205	NaN	NaN	140.741660	140.675439	2.354558	6.911213	104.866444	348.997893	112.934207

Exercise 2: Discovering the Pandas Verbs for Data Manipulation

First, let's get familiar with operations on columns.

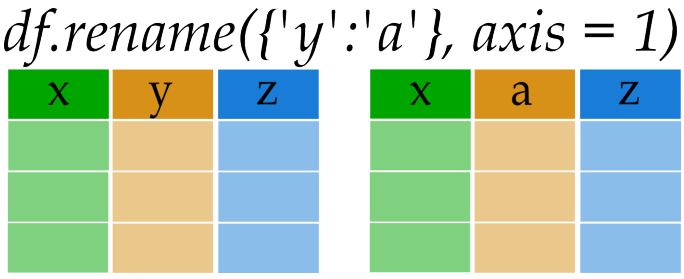
1. Create a `DataFrame` `emissions_copy` keeping only the columns `INSEE commune`, `Commune`, `Autres transports`, and `Autres transports international`.

Hint for this question



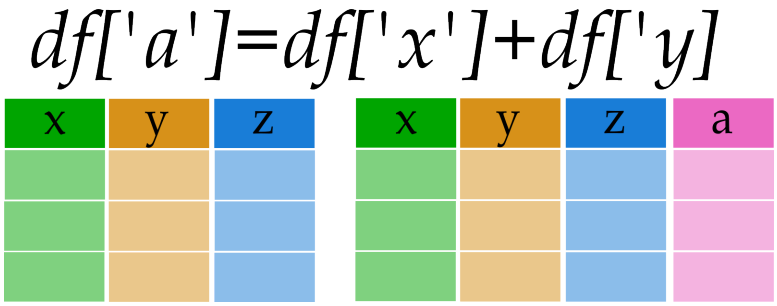
2. Since the variable names are not practical, rename them as follows:
   - `INSEE commune` → `code_insee`
   - `Autres transports` → `transports`
   - `Autres transports international` → `transports_international`

Hint for this question



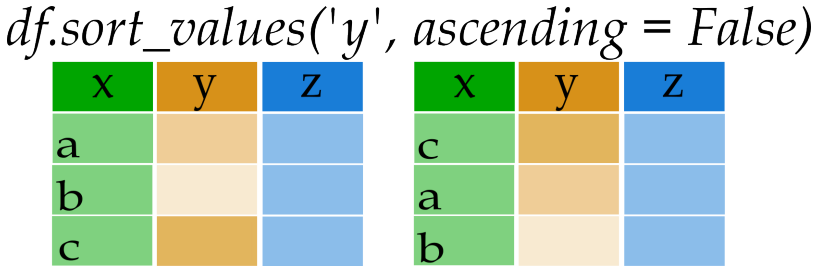
3. For simplicity, replace missing values (`NA`) with 0. Use the `fillna` method to transform missing values into 0.
4. Create the following variables:
   - `dep`: the department. This can be created using the first two characters of `code_insee` by applying the `str` method;
   - `transports_total`: the emissions of the transport sector (sum of the two variables).

Hint for this question



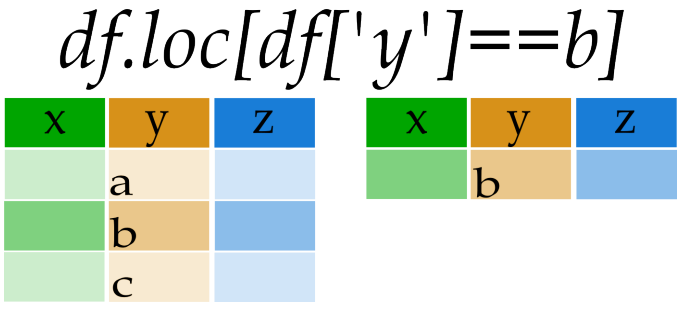
5. Order the data from the biggest polluter to the smallest, then order the data from the biggest polluter to the smallest by department (from 01 to 95).

Hint for this question



6. Keep only the municipalities belonging to departments 13 or 31. Order these municipalities from the biggest polluter to the smallest.

Hint for this question



Return to the initial `emission` dataset.

7. Calculate the total emissions by sector. Calculate the share of each sector in total emissions. Convert the volumes to tons before displaying the results.
8. Calculate the total emissions for each municipality after imputing missing values to 0. Keep the top 100 emitting municipalities. Calculate the share of each sector in this emission. Understand the factors that may explain this ranking.

Help if you are struggling with question 8 Play with the `axis` parameter when constructing an aggregate statistic.

In question 5, when the communes are ordered exclusively on the basis of the variable `transport_total`, the result is as follows:

	code_insee	Commune	transports	transports_international	dep	transports_total
31108	77291	LE MESNIL-AMELOT	133834.090767	3.303394e+06	77	3.437228e+06

	code_insee	Commune	transports	transports_international	dep	transports_total
31099	77282	MAUREGARD	133699.072712	3.303394e+06	77	3.437093e+06
31111	77294	MITRY-MORY	89815.529858	2.202275e+06	77	2.292090e+06

Question 6 gives us this classification:

	code_insee	Commune	transports	transports_international	dep	transports_total
4438	13096	SAINTES-MARIES-DE-LA-MER	271182.758578	0.000000	13	271182.758578
4397	13054	MARIGNANE	245375.418650	527360.799265	13	772736.217915
11684	31069	BLAGNAC	210157.688544	403717.366279	31	613875.054823

In question 7, the resulting table looks like this

	secteur	emissions	emissions (%)
2	transports_total	28774.0	50.0
1	transports_international	22239.0	39.0
0	transports	6535.0	11.0

	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire
4382	13039	FOS-SUR-MER	305.092893	1893.383189	1.722723e+04	50891.367548	275500.374439	2.296711e+06	6.765119e+06	9466.388806	74631.401993	42068.140058
22671	59183	DUNKERQUE	811.390947	3859.548994	3.327586e+05	71922.181764	23851.780482	1.934988e+06	5.997333e+06	113441.727216	94337.865738	70245.678455
4398	13056	MARTIGUES	855.299300	2712.749275	3.043476e+04	35925.561051	44597.426397	1.363402e+06	2.380185e+06	22530.797276	84624.862481	44394.822725
30560	76476	PORT-JEROME-SUR-SEINE	2736.931327	121.160849	2.086403e+04	22846.964780	78.941581	1.570236e+06	2.005643e+06	21072.566129	9280.824961	15270.357772
31108	77291	LE MESNIL-AMELOT	782.183307	133834.090767	3.303394e+06	3330.404124	111.613197	8.240952e+02	2.418925e+03	1404.400153	11712.541682	13680.471909
...	...	...	...	...	...	...	...	...	...	...	...	...
30056	74281	THONON-LES-BAINS	382.591429	145.151526	0.000000e+00	144503.815623	21539.524775	2.776024e+03	1.942086e+05	44545.339492	25468.667604	19101.562585
33954	86194	POITIERS	2090.630127	20057.297403	2.183418e+04	95519.054766	13107.640015	5.771021e+03	1.693938e+04	110218.156827	91921.787626	73121.300634
1918	06019	BLAUSASC	194.382622	0.000000	0.000000e+00	978.840592	208.084792	2.260375e+02	4.362880e+05	538.816198	3949.838612	740.022785
27607	68253	OTTMARSHEIM	506.238288	679.598495	3.526748e+01	4396.853500	237.508651	1.172570e+03	4.067258e+05	3220.875864	19317.977050	4921.144781
28643	71076	CHALON-SUR-SAONE	1344.584721	1336.547546	6.558180e+01	77082.218324	570.662275	1.016933e+04	2.075650e+05	77330.230592	29020.646500	34997.262920

At the end of question 8, we better understand the factors that can explain high emissions at the municipal level. If we look at the top three emitting municipalities, we can see that they are cities with refineries:

	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	...	Part Autres transports	Part Autres transports international	Part CO2 biomasse hors-total	Part Déchets	Part Energie	Part Industrie hors-énergie	Part Résidentiel	Part Routier	Part Tertiaire	Part total
4382	13039	FOS-SUR-MER	305.092893	1893.383189	17227.254383	50891.367548	275500.374439	2.296711e+06	6.765119e+06	9466.388806	...	0.010860	0.188696	0.533799	2.889719	24.090160	70.959233	0.099295	0.782807	0.441252	100.0
22671	59183	DUNKERQUE	811.390947	3859.548994	332758.630269	71922.181764	23851.780482	1.934988e+06	5.997333e+06	113441.727216	...	0.044652	3.849791	0.832891	0.275949	22.386499	69.385067	1.312444	1.091425	0.812695	100.0
4398	13056	MARTIGUES	855.299300	2712.749275	30434.762405	35925.561051	44597.426397	1.363402e+06	2.380185e+06	22530.797276	...	0.067655	0.759035	0.895975	1.112249	34.082905	59.361219	0.561912	2.110523	1.107196	100.0

Thanks to our minimal explorations with `Pandas`, we see that this dataset provides information about the nature of the French productive fabric and the environmental consequences of certain activities.

References

- The site pandas.pydata serves as a reference
- The book `Modern Pandas` by Tom Augspurger: https://tomaugspurger.github.io/modern-1-intro.html

Informations additionnelles

> **i** Python environment
>
> This site was built automatically through a `Github` action using the `Quarto` reproducible publishing software (version 1.10.18).

The environment used to obtain the results is reproducible via uv. The `pyproject.toml` file used to build this environment is available on the `linogaliana/python-datascientist` repository

```
[project]
name = "python-datascientist"
version = "0.1.0"
description = "Source code for Lino Galiana's Python for data science course"
readme = "README.md"
requires-python = "≥3.13,<3.14"
dependencies = [
    "altair≥6.0.0",
    "cartiflette≥0.2",
    "contextily==1.6.2",
    "duckdb≥0.10.1",
    "folium≥0.19.6",
    "gdal==3.11.4",
    "graphviz==0.20.3",
    "great-tables≥0.12.0",
    "gt-extras≥0.0.8",
    "ipykernel≥6.29.5",
    "jupyter≥1.1.1",
    "jupyter-cache≥1.0.0",
    "kaleido≥0.2.1",
    "langchain-community≥0.3.27",
    "loguru==0.7.3",
    "markdown≥3.8",
    "nbclient≥0.10.0",
    "nbformat≥5.10.4",
    "nltk≥3.9.1",
    "openpyxl≥3.1.5",
    "pandas≥3.0",
    "pip≥25.1.1",
    "plotly≥6.1.2",
    "plotnine≥0.15",
    "polars≥1.8.2",
    "py7zr≥1.1.3",
    "pyarrow≥17.0.0",
    "pynsee≥0.1.8",
    "python-dotenv≥1.0.1",
    "python-frontmatter≥1.1.0",
    "pywaffle≥1.1.1",
    "requests≥2.32.3",
    "scikit-image≥0.24.0",
    "scikit-learn≥1.8.0",
    "scipy≥1.13.0",
    "seaborn≥0.13.2",
    "selenium<4.39.0",
    "spacy≥3.8.4",
    "webdriver-manager≥4.0.2",
    "wordcloud==1.9.3",
]

[tool.uv.sources]
gdal = [
  { index = "gdal-wheels", marker = "sys_platform == 'linux'" },
  { index = "geospatial_wheels", marker = "sys_platform == 'win32'" },
]
```

```
[[tool.uv.index]]
name = "geospatial_wheels"
url = "https://nathanjmcdougall.github.io/geospatial-wheels-index/"
explicit = true

[[tool.uv.index]]
name = "gdal-wheels"
url = "https://gitlab.com/api/v4/projects/61637378/packages/pypi/simple"
explicit = true

[dependency-groups]
dev = [
    "nb-clean>=4.0.1",
]
```

To use exactly the same environment (version of `Python` and *packages*), please refer to the documentation for `uv`.

Bibliography

[1] W. McKinney, *Python for data analysis: Data wrangling with Pandas, NumPy, and IPython.* " O'Reilly Media, Inc.", 2012.

[2] H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund, *R for data science.* " O'Reilly Media, Inc.", 2023.